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Outage Capacities and Spectral Efficiencies of Multiuser Detectors in a CDMA Cellular Environment

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ABSTRACT

A practically oriented transmission and decoding strategy is analyzed for a randomly spread direct-sequence code division multiple access (DS-CDMA) system with *slow* flat fading. A multi-cell model that adheres to Wyner's infinite linear cell-array setting is considered, where only adjacent-cell interference is present, and characterized by a *single* parameter $0 \le \alpha \le 1$ .It is assumed that all users employ equal rates and transmit powers. A *strongest-users decoding* scheme is considered where the receiver ranks the intra-cell users according to their received powers, and decodes only a subset of these users, equal in number to the largest integer for which decoding is successful, i.e., only those users with high enough received power. The *fraction of undecodable users* (FUU) is considered to be a system design parameter, and can be controlled by the equal rate employed by each of the users. The discussion is confined to asymptotic analysis, where both the number of users per cell and the processing gain go to infinity, while their ratio (referred to as the "system load") goes to some finite constant. System performance can be further optimized if the system load is a degree of freedom.

Four types of multiuser detection strategies are considered. The first is the "*conventional" matched-filter detector* that treats *all* (either intra-cell or inter-cell) interference as additive white Gaussian noise (AWGN). The second is the *linear MMSE detector* that knows the signature sequences of all interfering users (both intra-cell and in adjacent cells) and mitigates their interference by means of a linear MMSE filter. The third is a detector that "*optimally"* decodes the transmissions of the maximum decodable subset of intra-cell users, while taking into account the structure of the multiuser interference generated by all remaining undecodable *intra-cell* users (assuming their signatures are known at the receiver). *Out-of-cell interference* is treated as AWGN. This detector is referred to henceforth as the "single-cell optimum (SCO) detector". Finally, an "optimum" detector is considered (analogous to the SCO detector) that takes into account the structure of *all* undecodable multiuser interference (*both intra-cell and out-of-cell*), assuming all signatures are known at the receiver. This detector is referred to henceforth as the "single cell-site processing optimum (SCPO) detector".

The total capacities under outage constraint of these detection strategies, derived as functions of the FUU (equivalent to the outage probability), are analyzed and compared. The results show that a severe penalty is induced by the strongest-users decoding scheme in the minimum E_b / N_0 that enables reliable communications, as compared to $E_b / N_{0_{\min}} = \ln 2 \approx -1.59\,\mathrm{dB}$ (the corresponding value when all users are decoded, which requires a rate-adjustment feedback). The minimum penalty is $10\log_{10} e \approx 4.34\,\mathrm{dB}$ in Rayleigh fading channels, and it is attained by setting the FUU to $Q^* = 1 - 1/e$, the optimum choice at the low signal-to-noise ratio (SNR) regime. In a single-cell system (corresponding to taking $\alpha = 0$), it is shown that with a with a proper system design, the outage constrained capacity of the SCO or SCPO detectors (both coincide in a single-cell environment) approaches the optimum spectral efficiency asymptotically with E_b / N_0. When undecodable out-of-cell interference is introduced, all of the four multiuser detection strategies suffer a significant performance degradation for all E_b / N_0 values, as compared to their spectral efficiency (specifying the ultimate performance). This degradation emphasizes the crucial role of *rate-adjustment feedback* for all E_b / N_0 values in the presence of undecodable out-of-cell interference.